Advanced Data Science

Zuil Pirola

UNIVERSIDADE
LUSÓFONA

Understanding SVD (Singular Value Decomposition)

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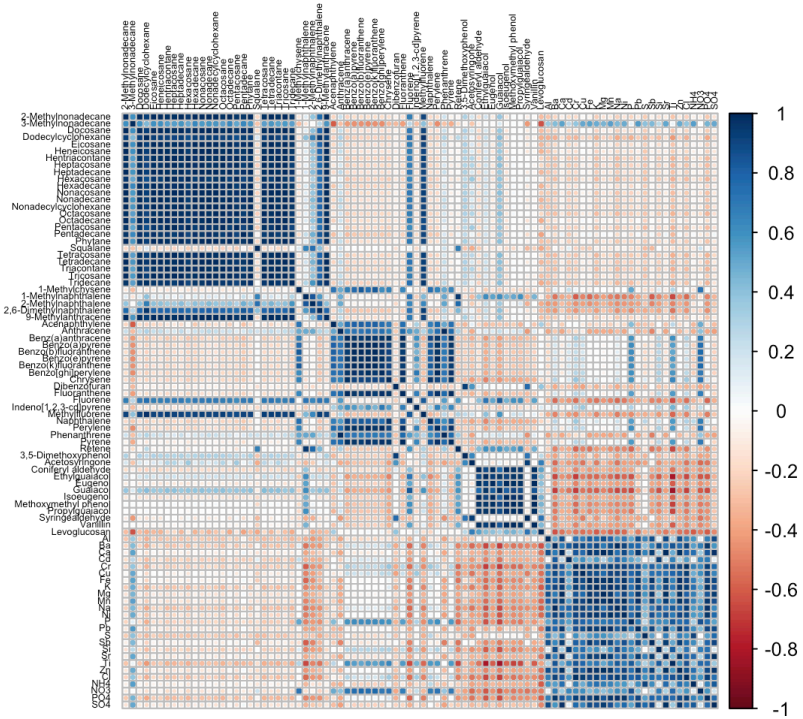
Why should we care?

- Real-world data looks complex
- But often has hidden structure
- Goal: find simple patterns inside complex data

Key Question

Are high-dimensional datasets really high-dimensional?

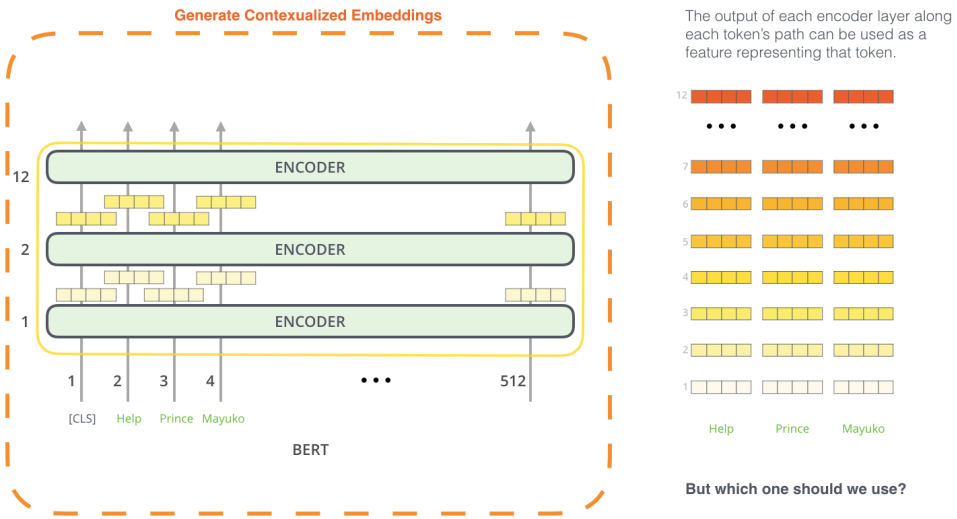


https://uncsrp.github.io/Data-Analysis-Training-Modules/high-dimensional-data-visualizations.html

Example

- Movie preferences
- Customer behavior
- Text data

Do these really need hundreds of dimensions?

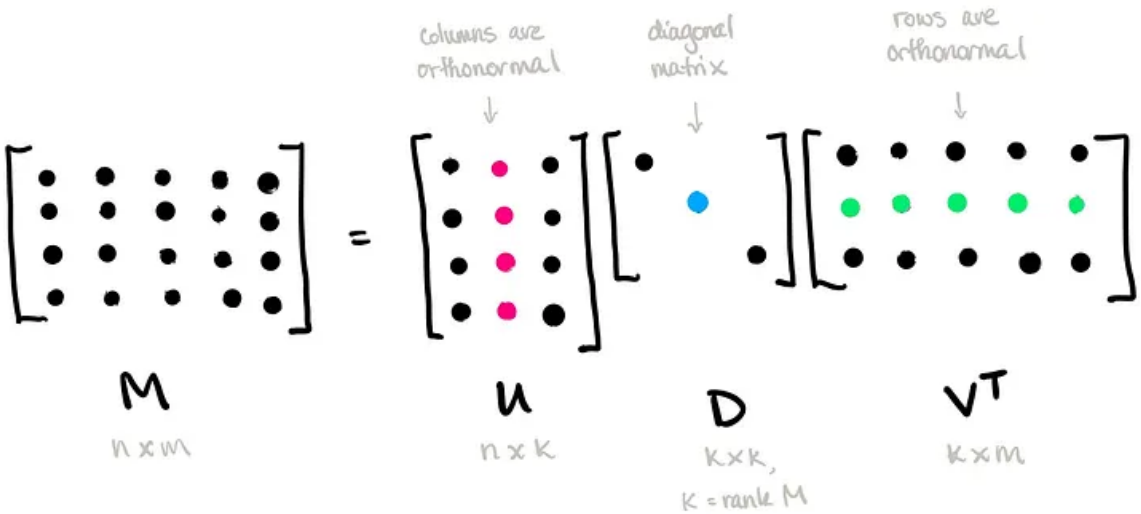


https://jalammar.github.io/illustrated-bert/

Core Idea

Data may live in a lower-dimensional structure

- Few underlying factors explain most variation



https://levelup.gitconnected.com/unlocking-insights-exploring-singular-value-decomposition-svd-and-its-dynamic-applications-238377ccbd51

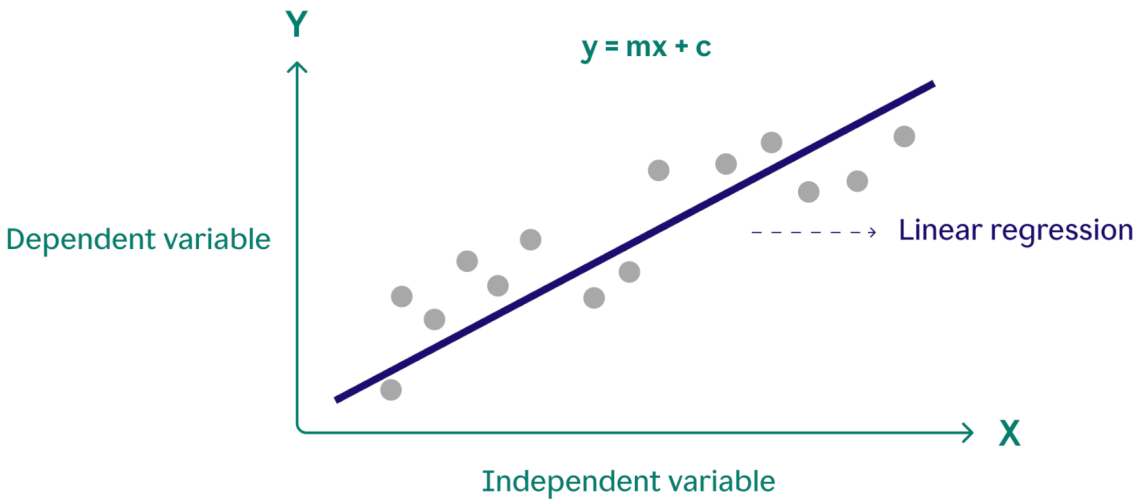
What does that mean?

- Many variables
- But highly correlated

Effective dimension is small

Visual intuition

Points in 2D but lying close to a line



Definition

Low-dimensional structure = few degrees of freedom explain the data

Quantos valores independentes precisas para descrever um ponto

Exemplos simples:

- Linha → 1 número
- Plano → 2 números
- Espaço 3D → 3 números

So how do we find these hidden directions?

SVD

SVD finds:

- Important directions in data
- Their importance

Data = sum of patterns

Data = $\text{pattern}_1 + \text{pattern}_2 + \text{pattern}_3 + \ldots$

SVD

Each pattern = direction in data
Each has a weight (importance)

Reorganizes data into:

- ordered patterns
- from most important to least

Direções → padrões de variação entre pontos

SVD sorts structure by importance

SVD

Considere uma matriz de users de um stream:
- Ação
- Romance
- Comédia

Identificar padrões como:
- gosta de ação
- gosta de romance
- gosta de filmes leves

SVD

SVD sorts structure by importance

Each pattern has a number:

how strong it is

Ex:
[100, 40, 10, 2, 0.5]

First few dominate

SVD

Ex:
[90, 10] - Valores singulares

Fator 1: Fator ação
Fator 2: Filmes leves

90 : quase tudo vem do fator ação (90%)

Energy = 90^2

Energy idea

Energy = how much information is captured

Big values = high energy

Most energy is concentrated in few components
How many components matter here?
[100, 40, 10, 2, 0.5]

We can ignore small components

Low-rank approximation

Keep only top k components

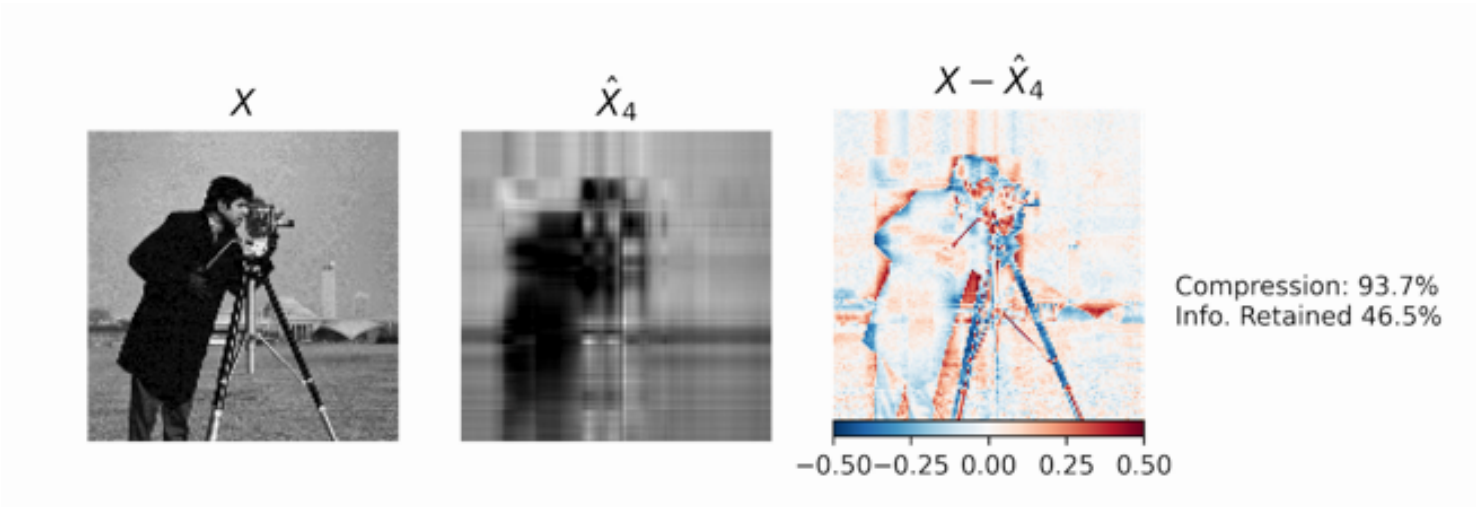
Low-rank approximation
Representar os dados usando só os padrões mais importantes

Original data ≈ simplified version

Best simple approximation of the data

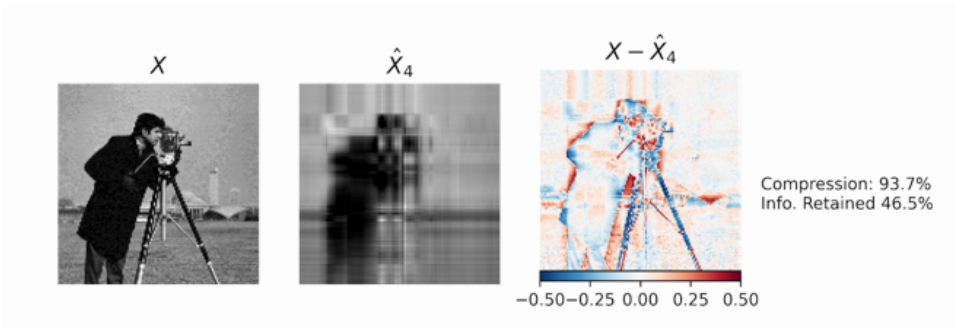
Analogy

- Image compression
- Keep main shapes
- Lose small details



https://dustinstansbury.github.io/theclevermachine/svd-data-compression

Analogy



- sem perda (compressão completa)
- com perda (compressão real)

$$A = U\Sigma V^T$$

What is removed?

Low values:
  Noise

Large singular values → signal
Small ones → noise

Real patterns repeat → strong
Noise is random → weak

Key takeaway

SVD separates structure from noise!!

Big picture

Data = structure + noise

SVD helps isolate structure

What if: Data is driven by hidden factors?

Ex:

- Taste (action vs romance)
- Economic forces
- Behavior patterns

Each data point = combination of factors

Summary

Few latent factors can explain complex data

- Data has hidden structure
- SVD finds ordered patterns
- Singular values measure importance
- Low-rank simplifies data
- Noise is filtered out

Final thought

Complex data is often simpler than it looks

- Where could this apply? (Your thesis?)
- What could be the hidden factors?

Obrigado!